

Y2 Electromagnetism 2

Non-assessed problem answers 2 (weeks 3–4)

Magnetostatics – EM induction – Dielectrics



1) Prove that the definition of the vector potential is satisfied, namely $\nabla \times \vec{A} = \vec{B}_0$ and $\nabla \cdot \vec{A} = 0$. A convenient reference frame is helpful, e.g. $\vec{B}_0 = B_0 \vec{e}_z$.

2) The magnetic field is directed horizontally,

$$B = \frac{\mu_0 i}{2} \left(\frac{r}{R} \right)^2.$$

3) The force is vertical:

$$F_z = \mu \cos \theta \cdot \frac{\partial B_z}{\partial z}.$$

4) The electromotive force is

$$\mathcal{E} = \frac{3\mu_0 m v r^2}{2L^4}.$$

5) The polarisability is $\alpha = 4\pi R^3$. For atomic hydrogen at STP,

$$\chi_E = \frac{p N_A}{RT} \cdot 4\pi r_{\text{Bohr}}^3 \approx 0.5 \times 10^{-4}.$$

$$6) \quad \vec{E}_2 = (-10\vec{e}_x + 20\vec{e}_y + 20\vec{e}_z) \text{ Vm}^{-1};$$

$$\vec{D}_2 = \varepsilon_0 \varepsilon_2 \vec{E}_2 = \varepsilon_0 (-30\vec{e}_x + 60\vec{e}_y + 60\vec{e}_z) \text{ Vm}^{-1};$$

$$\vec{P}_2 = \vec{D}_2 - \varepsilon_0 \vec{E}_2 = \varepsilon_0 (\varepsilon_2 - 1) \vec{E}_2 = \varepsilon_0 (-20\vec{e}_x + 40\vec{e}_y + 40\vec{e}_z) \text{ Vm}^{-1};$$

$$\sigma_p = \varepsilon_0 (E_{2z} - E_{1z}) = -10 \text{ C/m}^2.$$

$$7) \quad C = 39 \text{ pF}.$$

$$8) \quad U_1 = \frac{Q^2}{8\pi\varepsilon_0 R}; \quad U_2 = \frac{3}{5} \cdot \frac{Q^2}{4\pi\varepsilon_0 R}; \quad r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.8 \times 10^{-15} \text{ m}.$$

9) The energy release is 37% of the energy of uranium nucleus, i.e. 270 MeV.